

Using power amplifiers as high current buffers in the low Z VNA measurements

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Abstract—Accurately measuring ultra-low impedance, such as via resistance, presents a significant challenge, especially when using budget-friendly Vector Network Analyzers (VNAs). The primary limitation arises from the common current artifact, and the second comes from the noise floor, which both restrict the lowest impedance measurement possible. In this study, we introduce an enhanced measurement setup, referred to as Turbo VNA, which incorporates a ferrite chokes and an RF amplifier to extend the VNA's sensitivity. This approach effectively lowers the noise floor, allowing impedance measurements down to $8 \mu\Omega$, significantly improving upon the $250 \mu\Omega$ limitation of a standard VNA.

Index Terms—Low impedance, Vector Network Analyzer (VNA), Turbo VNA, noise floor, Ferrite chokes, RF amplifier, pre-amplification, via resistance, S21 response, power distribution network (PDN).

I. INTRODUCTION

As clock frequencies increase and signal switching intensifies, managing switching noise on the power distribution network (PDN) becomes crucial for ensuring reliable system performance. A well-designed PDN must maintain low impedance across its operating frequency range to minimize voltage fluctuations, reduce electromagnetic interference (EMI), and ensure stable power delivery to high-speed circuits. However, accurately measuring ultra-low impedance, such as via resistance and PDN characteristics, presents significant challenges due to the limitations of standard measurement equipment, particularly budget Vector Network Analyzers (VNAs) with high noise floors.

The primary goal of this project is to explore cost-effective methods for ultra-low impedance measurement using a budget VNA enhanced with pre-amplification. Specifically, we aim to:

- Evaluate the limitations of standard VNAs in measuring low impedances.
- Develop an improved measurement approach, referred to as Turbo VNA, by integrating ferrite chokes and an RF amplifier to extend the effective noise floor.
- Characterize the accuracy and reliability of this method by comparing results with hacked models.

By addressing these challenges, we seek to determine whether an affordable VNA setup can provide a viable alternative to high-end measurement systems for low-impedance characterization.

WORKING PRINCIPLE

The most accurate method for measuring DC resistance is the 4-wire Kelvin method, which eliminates lead and contact resistance errors. However, for AC resistance measurements, the 2-port shunt method is preferred over 1 1-port measurement.

In the 2-port shunt method, as illustrated in Figure 1 below, the Device Under Test (DUT) is connected between two ports of a Vector Network Analyzer (VNA). The resistance is determined by measuring the voltage drop across the DUT, which is observed at Port 2. The measured S21 is used to calculate the impedance, which ensures precise impedance characterization across a wide frequency range.

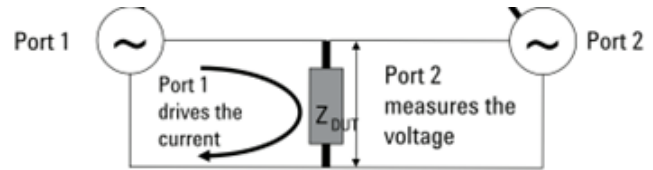


Fig. 1. 2-port shunt schematic

$$Z_o = 25 * \frac{S21}{(1 - S21)}$$

Fig. 2. Expression to derive Impedance from S21 response

LIMITATION

Every Vector Network Analyzer (VNA) port has a minimum detectable signal level, corresponding to the noise floor. This noise floor represents the lowest signal strength the VNA can accurately measure.

When measuring impedance, if the voltage drop across the Device Under Test (DUT) is significantly lower than the VNA's noise floor, the measurement will be limited by the noise floor itself. In such cases, instead of capturing the actual low impedance of the DUT, the VNA will only report the impedance corresponding to the noise floor, leading to inaccurate results. The challenge becomes particularly significant when measuring very low resistance, such as via resistance, as

it often falls below the VNA's detection threshold. As shown in Figure 3:

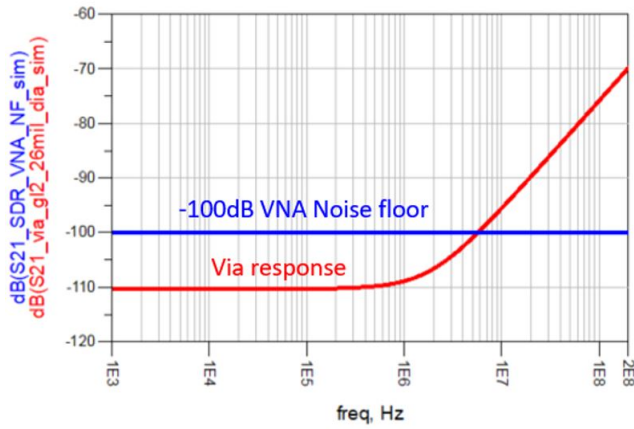


Fig. 3. VNA Noise floor(simulated) vs Via response(simulated)

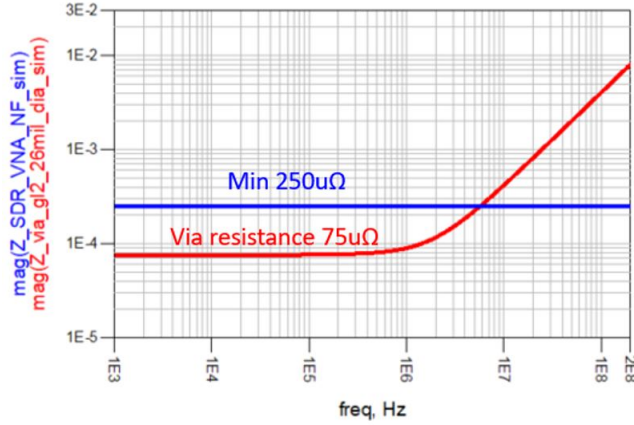


Fig. 4. Impedance - VNA Noise floor(simulated) vs Via(simulated)

- Fig 3 indicates that the SDR VNA's noise floor is approximately -100 dB, while the voltage drop across the via is around -110 dB.
- With this -100dB noise floor, the VNA can reliably measure impedances as low as 250 $\mu\Omega$.
- However, voltage drop across via is -110dB, which is well below -100dB noise floor of the VNA, the measurement would be limited by the VNA's sensitivity.
- As a result, instead of capturing the actual 75 $\mu\Omega$ impedance, the VNA would incorrectly report 250 $\mu\Omega$.

This highlights the importance of using a VNA with a lower noise floor or alternative measurement techniques for accurately characterizing ultra-low impedance components.

APPROACH

To overcome the noise floor limitation in budget VNAs, we developed a methodology that involves pre-amplifying the signal before it reaches Port 2. By amplifying the DUT's response, we ensure that the measured signal remains above

the VNA's noise floor, allowing for more accurate impedance characterization.

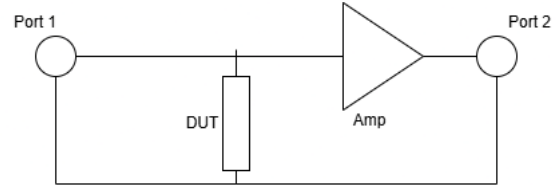


Fig. 5. 2-port shunt with preamp schematic

After capturing the pre-amplified response, we use a SPICE simulation tool to rescale the vector form of the S-parameter response to an unamplified value. This approach effectively compensates for the noise floor limitation.

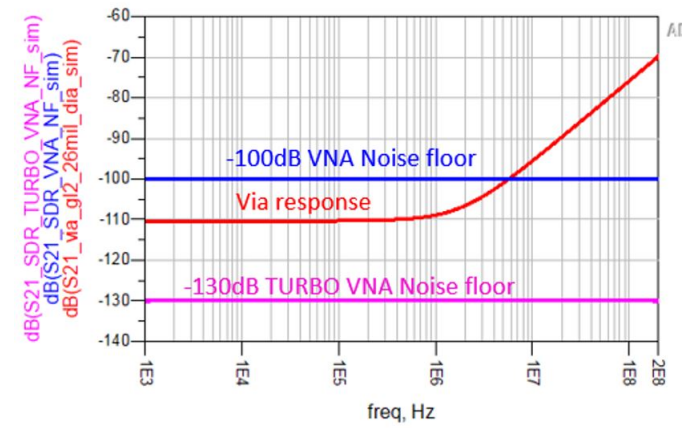


Fig. 6. VNA Noise floor(sim) vs Via response(sim) vs Turbo VNA Noise floor(sim)

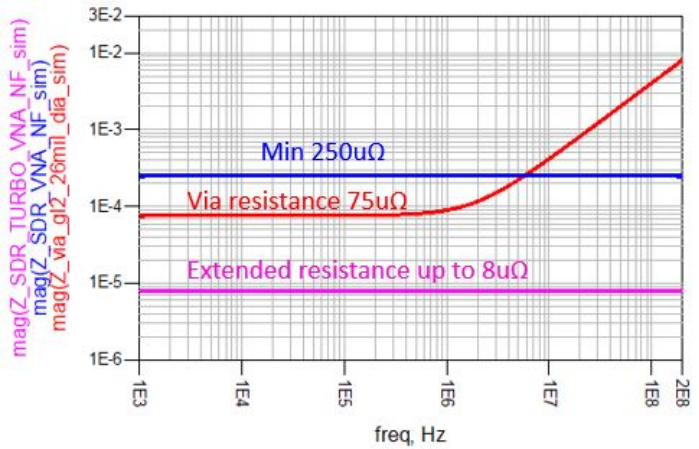


Fig. 7. Impedance - VNA Noise floor(sim) vs Via(sim) vs Turbo VNA noise floor(sim)

The VNA setup with pre-amplification is referred to as Turbo VNA. In Figure 6 the Turbo VNA setup enhances the measurement capabilities of budget VNAs by incorporating a +30 dB gain amplifier before Port 2. This pre-amplification

ensures that even extremely small voltage drops across the DUT are amplified before reaching the VNA, allowing for more accurate low-impedance measurements. For instance, if a DUT produces a voltage drop of -120 dB, the amplifier boosts this signal by +30 dB, making the effective voltage magnitude at Port 2 be -90 dB. Despite the VNA still reading -90 dB, the actual measurement corresponds to -120 dB, effectively lowering the noise floor of the system. This improvement allows for impedance measurements as low as $8 \mu\Omega$, compared to the $250 \mu\Omega$ limit of a standard setup. By extending the VNA's sensitivity without requiring expensive equipment, Turbo VNA provides an efficient and cost-effective solution for ultra-low impedance characterization.

SELECTING AN AMPLIFIER

For this low-cost, low-impedance measurement application, it was essential to select an amplifier that is both affordable and effective. To achieve this, need to carefully handpick multiple RF amplifiers that are readily available. However, sometimes amplifiers lack datasheets, requiring us to characterize them manually before determining their suitability for this application. Among the tested amplifiers, one of the best-performing options is shown in Fig. 8 and its Bode plot in Fig. 9, featuring a gain of +31 dB.



Fig. 8. RF amplifier

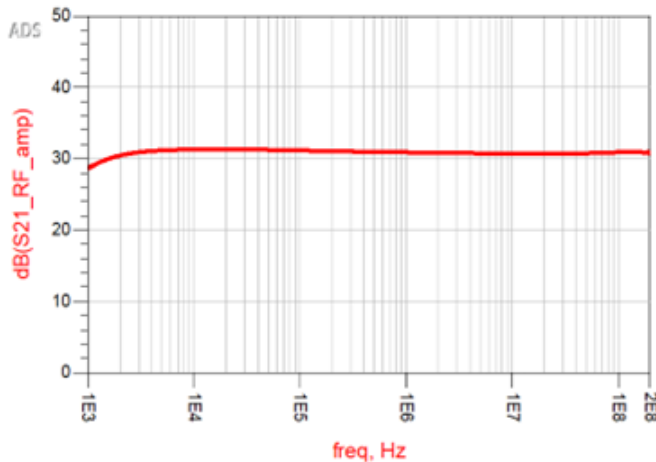


Fig. 9. Bode plot of RF amplifier

NOISE FLOOR

Before performing any measurements, it is crucial to understand the capabilities and limitations of the measuring device. In our case, the budget SDR kit VNA has the ability to sweep frequencies from 1 kHz (close to DC) up to 1.3 GHz. Given our objective to measure impedance at lower frequencies and the variable noise floor behavior of the VNA, we can focus on a frequency range from 1 kHz to 200 MHz for both measurement and modeling. The measured noise floor for both the standard VNA and the Turbo VNA setup is shown in Fig. 10, providing insight into the improved sensitivity achieved by adding ferrite chokes and the pre-amplification approach.

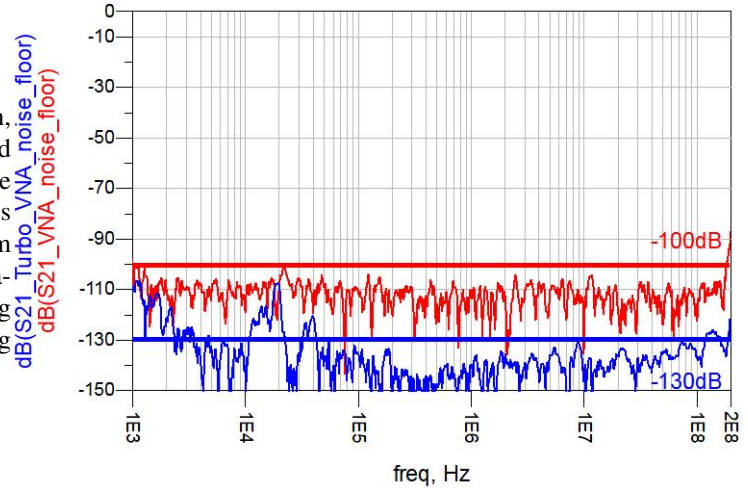


Fig. 10. VNA noise floor(meas) vs Turbo VNA noise floor(meas)

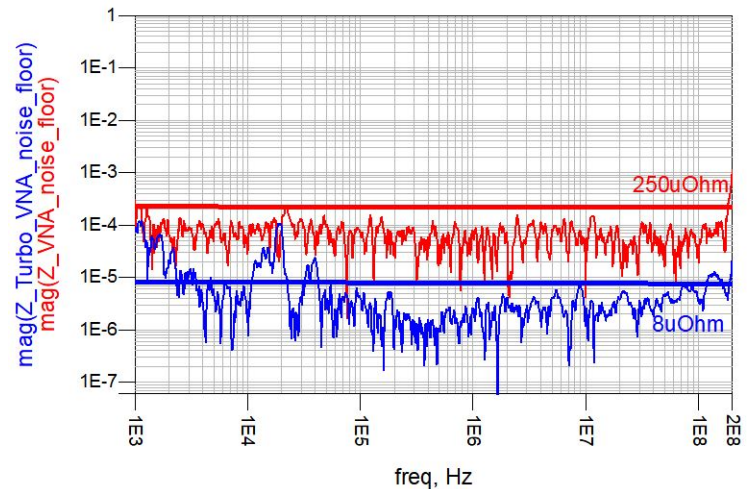


Fig. 11. Impedance comparison

According to the user guide of the VNA, the noise floor in the specified bandwidth is typically -90 dB. However, based on our measured response, we found that the VNA can reliably measure up to -100 dB without introducing errors or data loss, which corresponds to an impedance of $250 \mu\Omega$. When we added the RF amplifier, the noise floor was further lowered

to -130 dB without compromising the accuracy of the data. As a result, the VNA's impedance measurement capability improved, enabling us to measure down to an impressive $8 \mu\Omega$.

CONSISTENCY CHECK

To ensure that the Turbo VNA setup and measuring system are functioning correctly, it's essential to verify the accuracy of measurements by testing a known component with a well-understood behavior. For this purpose, we measured a $10\text{m}\Omega$ resistor mounted on the test board. This provides a reference point to compare the 4-wire Kelvin method, the normal VNA, the Turbo VNA's measurements, and simulated results with respect to the noise floor. But before that, we were

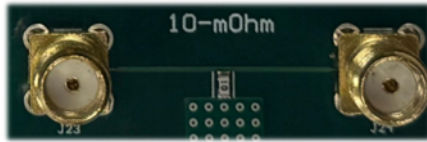


Fig. 12. $10\text{m}\Omega$ structure

surprised by the common current artifact shown in Figure 13, where we observed that this unwanted gain at the lower frequency in S-parameters led to a high resistance value at the lower frequency, as shown in the red trace, instead of a flat resistance response. To avoid this common current artifact,

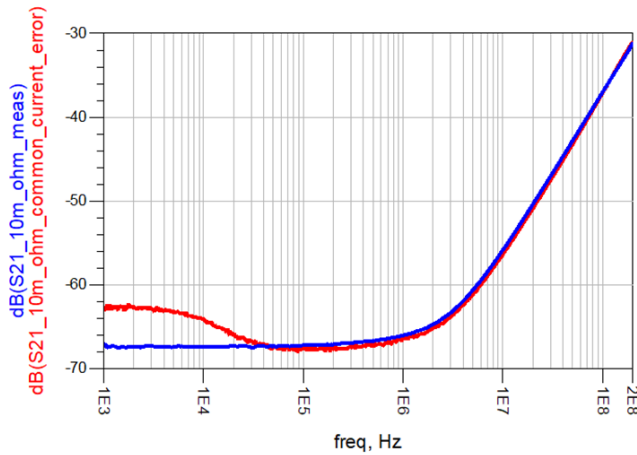


Fig. 13. Common current artifact

ferrite chokes were added near port-2 as shown in Figure 14 reduced the common current noise and enabled us to see a flat resistance response, as shown in the blue trace. After that, we performed measurements using both the Normal and Turbo VNA setups. The actual resistance obtained from the Normal setup measurements was approximately $10.65\text{m}\Omega$, determined by simulating the resistance using the ADS RL model and minimizing the uncertainty to $0.04\text{m}\Omega$, as indicated by the similarity metric shown in Figure 17. The inferred resistance values from both setups showed only a 0.03% to 27% variation



Fig. 14. Turbo VNA Setup

relative to the impedance value of $10.618\text{m}\Omega$, which was independently confirmed against the DC 4-wire method. This small discrepancy in the Normal VNA setup can be attributed to the inherent differences between the measurement systems and the characteristics of the device under test, factors that are typical in ultra-low impedance measurement scenarios.

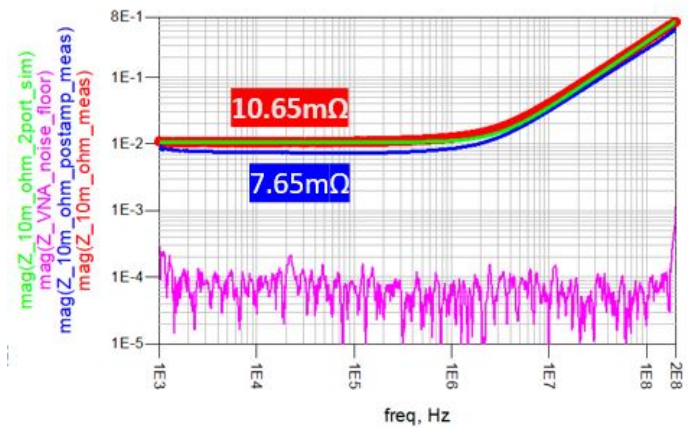


Fig. 15. $10\text{m}\Omega$ Impedance comparison

Nominal	DC 4 wire method	Normal VNA	Turbo VNA
$10\text{m}\Omega \pm 1\%$	$10.618\text{m}\Omega$	$10.65 \pm 0.04\text{m}\Omega$	$7.675 \pm 2.9\text{m}\Omega$

TABLE I
MEASURED DATA COMPARISON

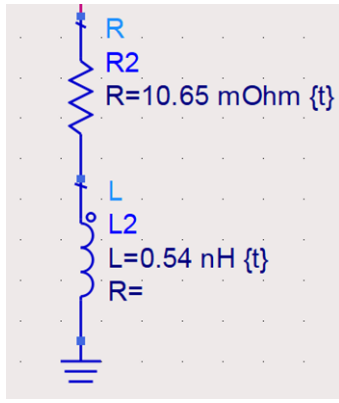


Fig. 16. 10m Ω Resistor RL model from ADS

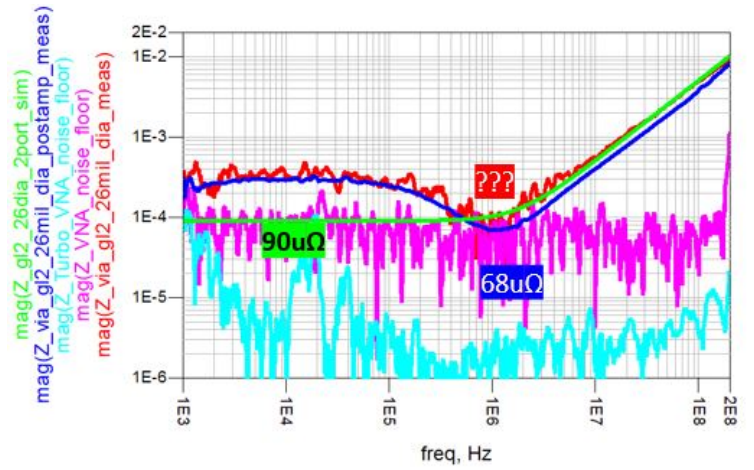


Fig. 19. Via impedance(diameter = 26mil, length = 8mil)

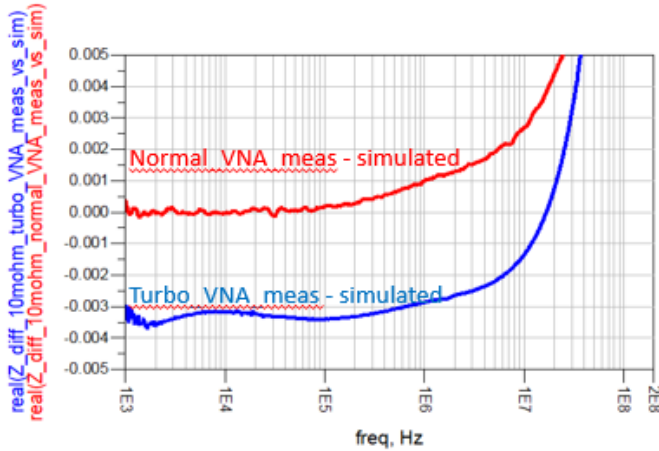


Fig. 17. Similarity Metric

ULTRA-LOW IMPEDANCE MEASUREMENT (VIA RESISTANCE)

Finally, we attempted to measure ultra-low impedance, specifically the via resistance. The S21 response of the via is closer to the noise floor of the normal VNA, making it difficult to obtain an accurate measurement. As a first-order



Fig. 18. Via structure

estimation, we applied a rule of thumb to approximate the via resistance, which resulted in a value of 56 $\mu\Omega$ well below the detection threshold of the normal VNA. Using the Turbo VNA response, identified that at the dip where both inductance slope and resistance flat response will intersect, at

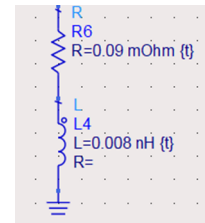


Fig. 20. Via model from ADS

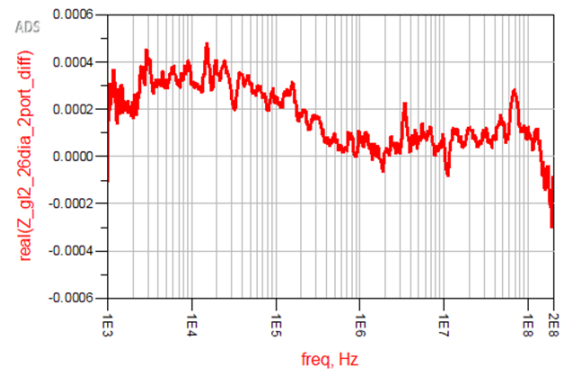


Fig. 21. Similarity metric(diameter = 26mil, length = 8mil)

This confirms that the Turbo VNA methodology helps measure ultra-low impedance components well below normal

First order	ADS Hacked	DC 4 wire	Normal VNA	Turbo VNA
56 $\mu\Omega$	90 $\mu\Omega$	346 $\mu\Omega$	NA	68 $\pm$ 22 $\mu\Omega$

TABLE II
MEASURED DATA COMPARISON

VNA operation, making it a reliable alternative to high-end VNAs for low Z measurements.

CONCLUSION

This study demonstrates that ultra-low impedance measurements down to approximately 8 $\mu\Omega$ can be reliably achieved using a cost-effective VNA in combination with an RF amplifier. By carefully characterizing the amplifier's input/output impedance and wideband gain, we ensured suitability for precision low-impedance measurements, even in the absence of detailed datasheet specifications.

Key limitations typically encountered with low-cost VNAs were addressed through practical hardware enhancements. The use of ferrite chokes mitigated common current artifacts, reducing measured impedance from 14m Ω to 0.45m Ω .

Despite the inherent gain rise at low frequencies, the structure's resistance, approximately 90 $\mu\Omega$ was estimated as 68 $\mu\Omega$ with an uncertainty of 24%, a significant improvement over first-order approximations and measurements from the Normal VNA setup. Overall, this approach validates the Turbo VNA methodology as a practical, low-cost solution for accurate ultra-low impedance characterization.

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REFERENCES

- [1] 2-Port shunt Method explanation. <https://www.keysight.com/th/en/assets/7018-08474/application-notes/5989-5935.pdf>
- [2] RF amps information. <https://www.alibaba.com/product-detail/RF-Wideband-Amplifier-30dB-Low-noise-1600090642908.html>
- [3] SDR Kits Website. <https://www.sdr-kits.net/index.php?route=common/home>
- [4] Novak, Istvan. "Why 2-Port Low-Impedance Measurements Still Matter." Signal Integrity Journal, January 2020. <https://www.signalintegrityjournal.com/articles/1544-why-2-port-low-impedance-measurements-still-matter>
- [5] Eric Bogatin, Sakshi Ranjan, Yash Shah, Jeff Cain, and Joe Hock "A Low-Cost Capacitor Characterization System". <https://www.signalintegrityjournal.com/articles/1976-a-low-cost-capacitor-characterization-system>
- [6] Variation response due to ground loop. <https://www.picotest.com/wp-content/uploads/2024/06/Ultra-low-Impedance-20-micro-ohm-Measurement-using.pdf>
- [7] Phase error discussion. <https://groups.io/g/VNWA/topic/74472949>
- [8] The 2-Port Shunt-Thru Measurement and the Inherent Ground Loop. <https://www.signalintegrityjournal.com/articles/1202-the-2-port-shunt-thru-measurement-and-the-inherent-ground-loop>

- [9] The Challenge of Measuring a 40 $\mu\Omega$, 2000 Amp PDN with a 2-Port Probe: How Much CMRR is Needed? <https://www.signalintegrityjournal.com/blogs/15-extreme-measurements/post/3378-the-challenge-of-measuring-a-40-u-2000-amp-pdn-with-a-2-port-probe-how-much-cmrr-is-needed>